LAB 2 WIRESHARK封包觀察

OUTLINE

- 1. introduction
  - 實驗目的
  - Wireshark 簡介 and Installation
- 2. Lab 2-1:觀察Wireshark封包
- 3. Lab 2-2:截取帳密
- 4. Exercises:
  - Lab 2-1, Lab 2-2
  - Note: Your jobs
    - Please upload your Lab Exercise results to the Tronclass couse website (one file per group)

實驗目的

- 了解網路協定(network protocol)在網路環境中所扮演的角色
- 透過實際觀察封包的組成及特定協定的運作，了解通訊協定層和網路運作機制。
- 熟悉網路封包分析軟體(WireShark)的操作流程。

WIRESHARK簡介

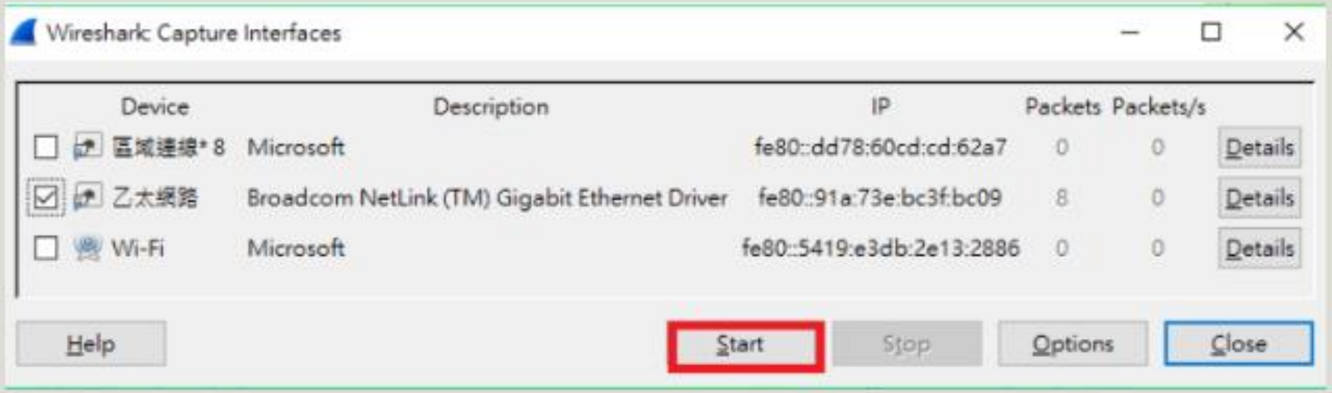
- Wireshark 是免費開源的網路封包分析軟體，網路封包分析軟體的功能是截取網路封包，並盡可能顯示出最為詳細的網路封包資料。
- 如果電腦沒有WireShark軟體，可至https://www.wireshark.org/download.html下載

WIRESHARK 介面使用介紹

➤ 操作

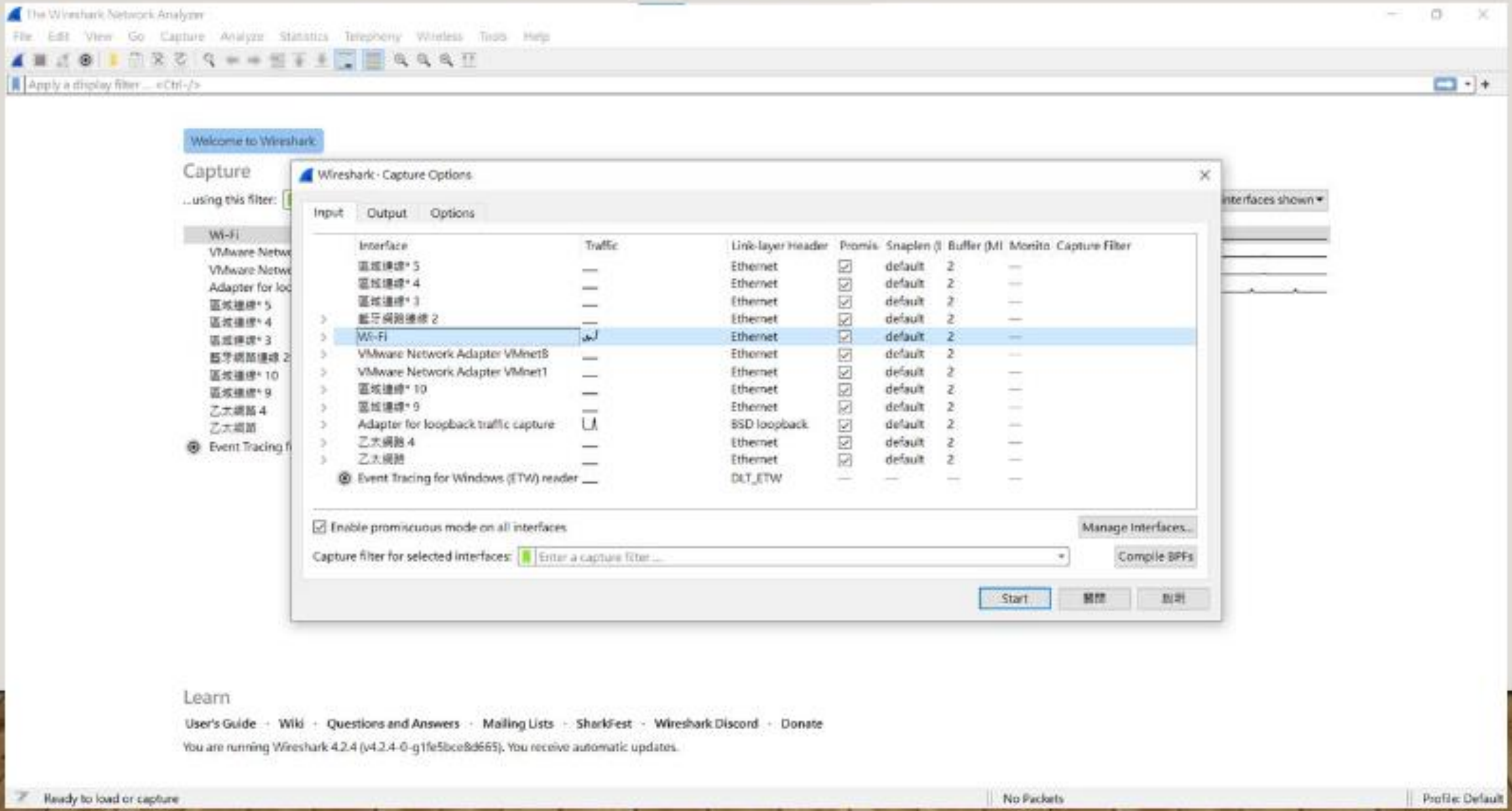
- 執行、接收封包資料

- 暫停、看目前封包值



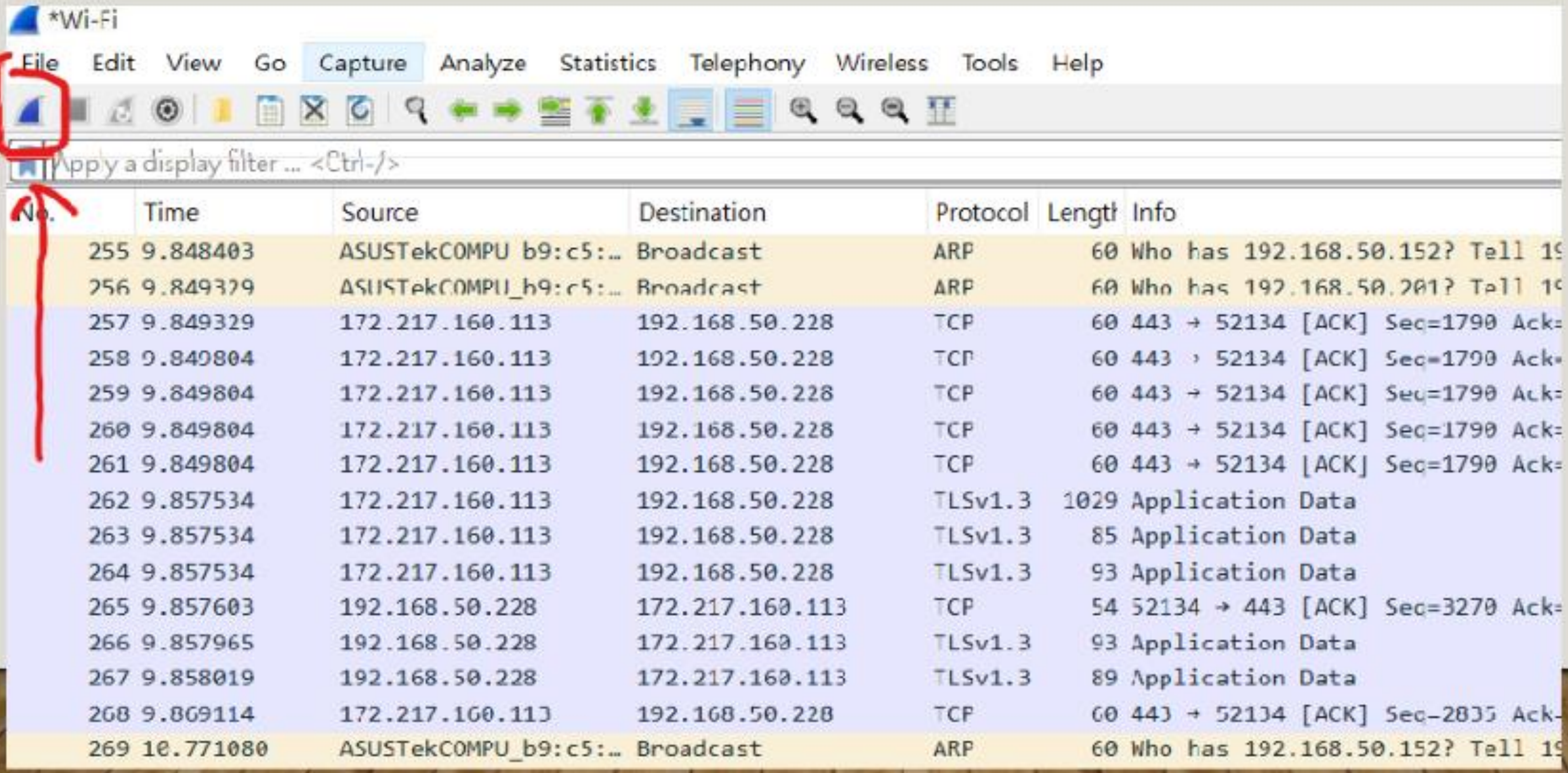
選擇網路

- 點選capture以選取合適的網路 註:範例是用Wi-Fi



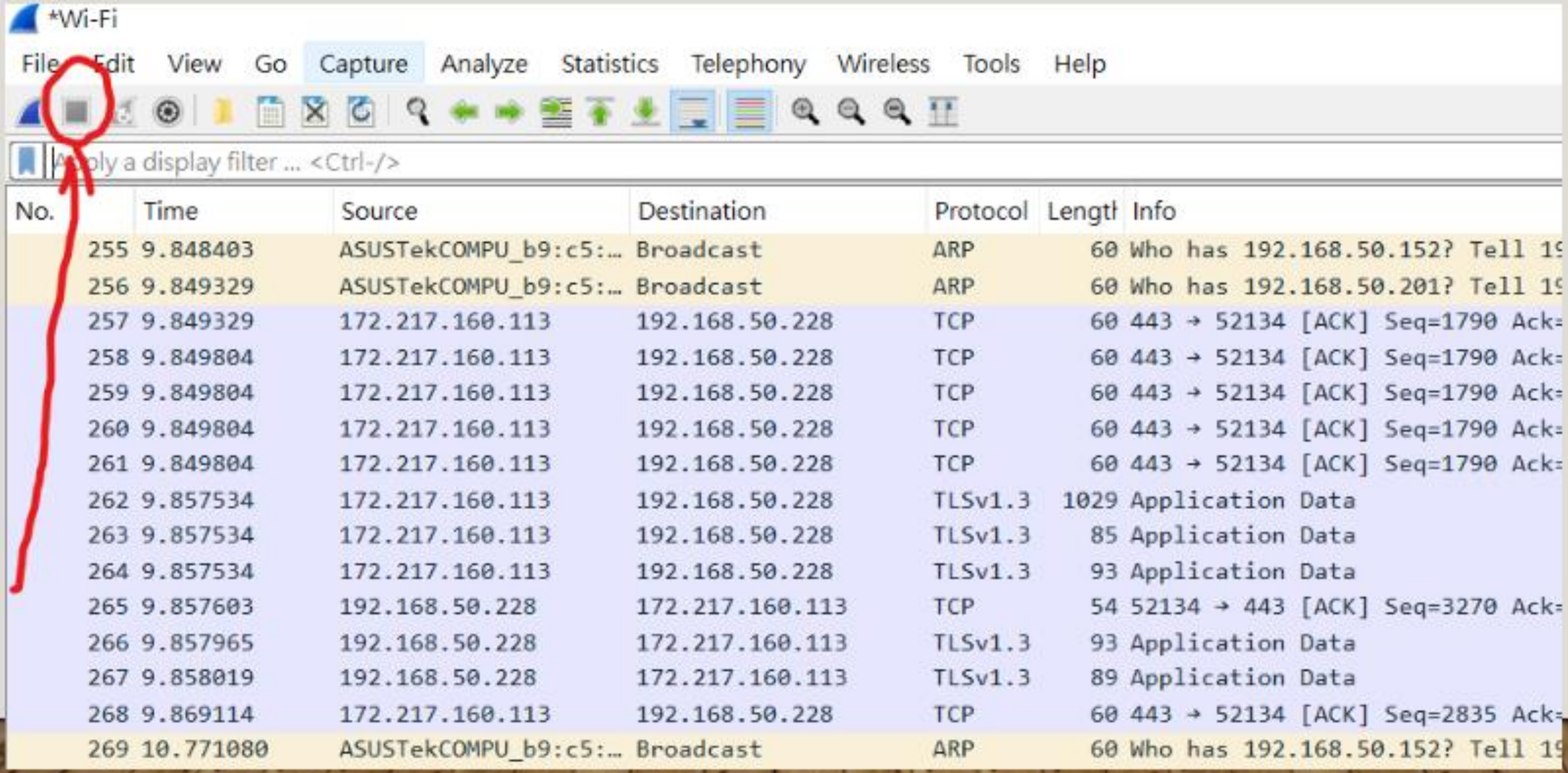
開啟封包接收

- 點選左上的魚鰭狀圖案，開始執行封包接收



暫停封包接收

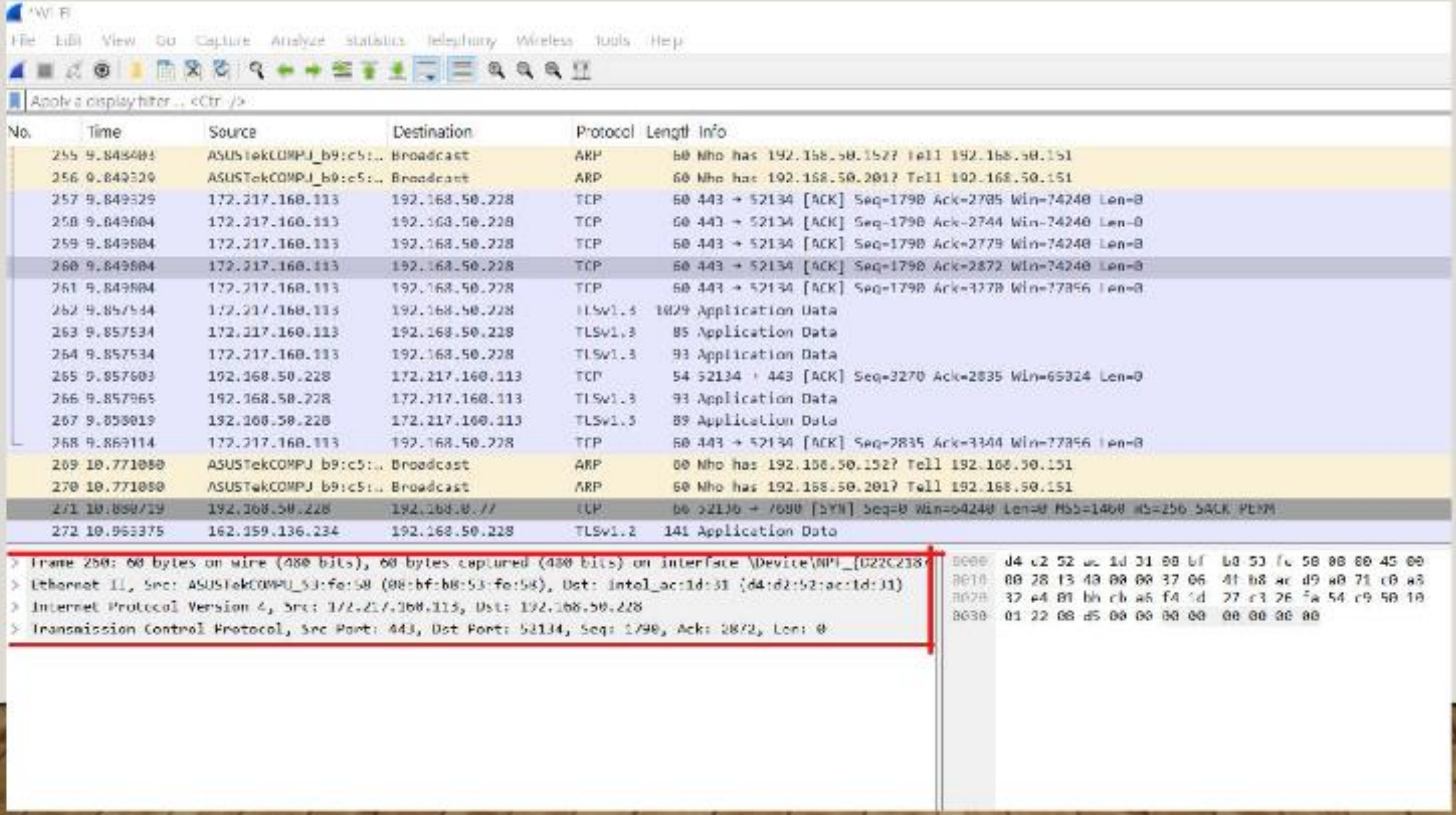
- 點選左上方正方型圖案暫停封包接收



No.	Time	Source	Destination	Protocol	Lengtl	Info
255	9.848403	ASUSTekCOMPU_b9:c5:…	Broadcast	ARP	60	Who has 192.168.50.152? Tell 1…
256	9.849329	ASUSTekCOMPU_b9:c5:…	Broadcast	ARP	60	Who has 192.168.50.201? Tell 1…
257	9.849329	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack…
258	9.849804	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack…
259	9.849804	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack…
260	9.849804	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack…
261	9.849804	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack…
262	9.857534	172.217.160.113	192.168.50.228	TLSv1.3	1029	Application Data
263	9.857534	172.217.160.113	192.168.50.228	TLSv1.3	85	Application Data
264	9.857534	172.217.160.113	192.168.50.228	TLSv1.3	93	Application Data
265	9.857603	192.168.50.228	172.217.160.113	TCP	54	52134 → 443 [ACK] Seq=3270 Ack…
266	9.857965	192.168.50.228	172.217.160.113	TLSv1.3	93	Application Data
267	9.858019	192.168.50.228	172.217.160.113	TLSv1.3	89	Application Data
268	9.869114	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=2835 Ack…
269	10.771080	ASUSTekCOMPU_b9:c5:…	Broadcast	ARP	60	Who has 192.168.50.152? Tell 1…

查看封包內容

- 以下(紅色框內)可以查看封包的詳細內容



作業繳交

- Wireshark 頁面必須含有TCP封包，並截圖